Agreement Attraction Meets Negative Concord: Insights from Speeded Choice and Comprehension

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Aim

- Agreement attraction occurs when a verb agrees with an intervening noun instead of the subject (Bock & Miller 1991, Wagers et al. 2009, etc.):

The key to the cabinets are on the table

- Agreement attraction has always been tested in positive statements and with positive intervenors.
- The role of negation in attraction remains unclear.
- Languages differ in how they express negation (Zeijlstra 2004): negative concord/ no negative concord).

Question: Do negative intervenors trigger agreement attraction in negative sentences in Romanian?

Cheia de la niciunele dintre cabinete nu sunt pe masă
key.~the of to none of cabinets not are on table
The key to none of the cabinets are not on the table'.

Why Romanian?

Agreement attraction

- Romanian shows a small agreement attraction effect in locative structures: BNs < DPs (Bleotu & Dillon 2024)
- **Locative Determiner Omission:** The definite article is dropped after locative prepositions. (Prendergast 2007)
- **The Naked Noun Constraint**: An unmodified common noun in the preverbal position cannot be the surface subject of a sentence under conditions of normal stress and intonation. (Dobrovie-Sorin 2013).

=> BN intervenors are not subject-like

BUT negative quantifiers are subject-like!

Negative concord

- Romanian is a **strict negative concord language.**
- If we have a negative quantifier, negation has to also be expressed on the verb.

N.B. The negations do not cancel each other out, they represent **a single negation.**

- **Negative Concord** = "the phenomenon during which "the negative feature is spread (…) over all elements in the sentence (or the clause) that can bear it" (Van de Wouden, 1944, p. 93)
- A negative quantifier subject requires negation on the verb => if the speaker erroneously thinks the subject is negatively quantified (either because they take the negative intervenor as the subject or because they spread the negation and agreement features onto the subject), they are more likely to persist in that error if they encounter a negation on the verb.

Predictions

- Negative quantifiers are subject-like, and thus we expect them to induce attraction.
- Under **cue-based retrieval theories** (Wagers et al. 2009, Dillon et al. 2013, Badeker & Kuminiak 2007), participants could take the intervenor as the subject (*niciunele dintre cabinete* 'none of the cabinets'), an erroneous possibility reinforced by the negative verb.
- Under **'representational' theories** (Eberhard et al. 2005, Keshev et al. 2025), several options arise:

(i) *only agreement percolation:* merely agreement features percolate to the head

i.e. the subject is *cheile* 'keys'

(ii) *agreement & negation percolation*: that both agreement and negation features percolate

i.e. the subject is *niciunele dintre chei* 'none of the keys'

N. B. *Only negation percolation* may be an option too.

- If participants take the subject to be negative (cue-based/ agreement & negation percolation), they are susceptible to a *negative concord illusion* (Lacina 2025) with a negative subject.

Participants

- 100 undergraduates (Romanian native speakers)

Procedure and materials

- a speeded forced choice continuation task (Staub 2009), where participants had to choose between a 3rd singular and a 3rd plural verbal form
- a comprehension task where they had to specify the subject of the sentence.

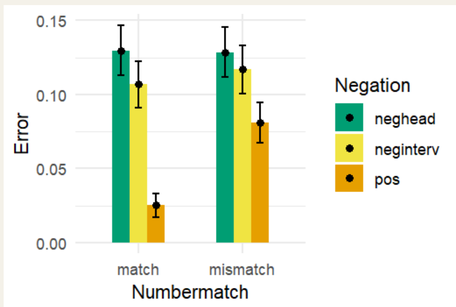
- 24 items with 6 conditions crossing:
  - ❑ **MATCH** (Number Match/ Mismatch) x
  - ❑ **NEGATION** (Positive (No negation inside the DP)/ Negative Head / Negative Intervenor).

- These were combined with 72 fillers.
- After each speeded critical item, participants were asked to choose the sentence subject from 8 choices :
  - ➢ SG Head
  - ➢ Negative SG Head
  - ➢ SG Intervenor
  - ➢ Negative SG Intervenor,
  - ➢ PL Head
  - ➢ Negative PL Head
  - ➢ PL Intervenor
  - ➢ Negative PL Intervenor

Examples of materials

Conditions exemplified	Example sentence
Number Match x Positive Head & Intervenor	Pisica de lângă fată nu au /are cat.the DE near girl not have.3pl/have.3sg
Number Match x Negative Intervenor	Pisica de lângă nicio fată nu au /are cat.the DE near no girl not have.3pl/have.3sg
Number Match x Negative Head	Nicio pisică de lângă fată nu au /are no cat DE near girl not have.3pl/have.3sg
Number Mismatch x Positive Head & Intervenor	Pisica de lângă fete nu au /are cat.the DE near girls not have.3pl/have.3sg
Number Mismatch x Negative Intervenor	Pisica de lângă niciunele dintre fetele nu au /are cat.the DE near none of girls.the not have.3pl/sg
Number Mismatch x Negative Head	Nicio pisică de lângă fete nu au /are no cat DE near girls not have.3pl/have.3sg
Comprehension Question: *Cine nu are (ceva)?* 'Who does not have something?' *pisica* 'cat.the', *nicio pisică* 'no cat', *fata* 'girl', *nicio fată* 'no girl, *pisicile* 'cats.the', *niciunele dintre pisici* 'none of cats.the', *fetele* 'girls.the', *niciunele dintre fete* 'none of girls.the'	

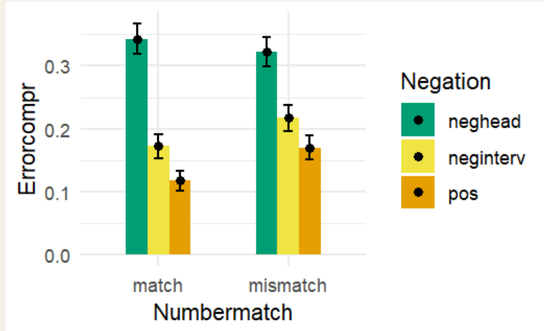
Results: Agreement



- Agreement errors were overall quite low (under 15%).
- We compared accuracy rates across conditions using a generalized linear mixed-effects model with agreement accuracy as the dependent variable, the fixed effects Numbermatch (match = +0.5, mismatch = −0.5), negation contrasts, and their interactions, random intercepts for participants and items, and corresponding random slopes.
- **Negation contrasts** used Helmert coding:
  - ❖ ContrastA: neginterv = 1 vs neghead = −1, pos = 0
  - ❖ ContrastB: neginterv = 1 vs pos = −1, neghead = 0
- Number Mismatch => more errors than Number Match.
- The main effect of Contrast A ($p < 0.01$) indicated more agreement errors in negative head structures than in negative intervenor structures in negative sentences.
- The interaction between Number Match and Contrast A was marginally significant ($p = 0.09$).
- We found a significant effect for Contrast B between negative and positive intervenors ($p < .01$), and a significant interaction between Match and Contrast B ($p = 0.01$): positive intervenors induce a significant difference between match and mismatch, unlike negative ones

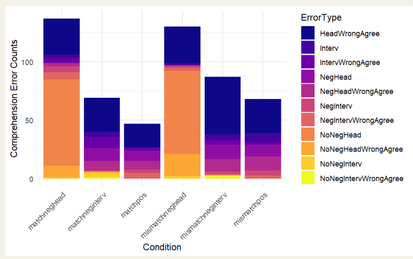
Parameter	Estimate	SE	z	*p*
Intercept	-2.886	0.172	-16.764	<.0001***
ContrastA (NegHead vs NegInterv)	-0.636	0.198	-3.204	0.001
Match	0.495	0.201	2.455	0.014
ContrastB (NegInterv vs Positive)	1.0315	0.2471	4.174	<.0001***
ContrastA:Match	0.414	0.251	1.652	0.098
ContrastB: Match	-0.706	0.296	-2.383	0.017

Results: Comprehension



Parameter	Estimate	SE	z	*p*
Intercept	-1.820	0.1611	-11.298	<.0001***
ContrastA (NegHead vs NegInterv)	-1.038	0.146	-7.100	<.0001***
Match	0.271	0.162	1.672	0.094.
ContrastB (NegInterv vs Positive)	0.811	0.1984	4.091	<.0001***
ContrastA:Match	0.406	0.189	2.158	0.03
ContrastB: Match	-0.271	0.250	-1.080	0.279

- The offline data reveals lower accuracy in comprehension vs. agreement. (Parker 2019, Keshev et al. 2025)
- A regression model parallel to the one for agreement reveals significant effects for ContrastA, ContrastB and the interaction between Contrast A and Match in comprehension.
- We see a variety of errors: agreement percolation errors (highest number), agreement and negation percolation errors, cue retrieval errors, negation drop, a.o.
- Some errors provide partial evidence for *negative concord illusions.*
- Some of these errors could be due to distributivity (Humphreys & Bocks 2005).
- N. B. We are currently doing an analysis removing errors that are semantically equivalent with correct answers (*Neg Head* = *NoNegHead NOT*).



Discussion

- **Several main findings:**

1. Negative intervenors lead to higher overall agreement and comprehension errors than positive ones.

➢ N. B. The error rates with negative intervenors are similar in Match and Mismatch.

2. Negative intervenors make the number agreement attraction effect found with positive intervenors vanish.

➢ This is contrary to our expectations of seeing more agreement attraction with negative intervenors.
➢ Not in line with cue-based retrieval and agreement only accounts

3. Positive Head-Negative Intervenors lead to fewer overall agreement errors than Negative Head-Positive Intervenors.

➢ percolation can potentially be downward as well.

=> Negative Concord properties interfere with attraction. The heterogenous error patterns- in line with agreement & negation percolation (Eberhard et al. 2005) or item-position binding (Keshev et al. 2025).

Acknowledgements

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